SECURITY INCIDENT REPORT

Executive Account Takeover Investigation

Detection Engineering & Response Assessment

Report Detail	Value
Organization	Cloudora (fictional / simulated)
Incident Type	Executive account takeover
Affected Account	CEO account — Daniel Reev
Priority	P1 — simulated
Investigation Platform	Splunk Enterprise
Telemetry Period	04 Aug 2026 – 11 Aug 2026
Data Type	Synthetic sign-in and audit telemetry
Report Status	Final

1. Executive Summary

A simulated P1 security incident was investigated after suspicious authentication activity was observed against the Cloudora CEO account. The investigation correlated sign-in and audit telemetry in Splunk to determine whether the activity represented an actual account compromise, identify actions performed after access, assess scope, and determine business impact.

The available evidence supports a confirmed compromise of the CEO account. The observed sequence was repeated failed authentication attempts, a successful authentication from the suspicious source, subsequent Outlook Web and Azure Portal access, registration of a new authenticator, and creation of an inbox rule affecting finance/invoice-related messages.

Assessment	Result
Account compromise	CONFIRMED
Password-spray activity	OBSERVED
Unauthorized security registration	CONFIRMED
Unauthorized mailbox rule	CONFIRMED
Unique accounts targeted	26
Successful suspicious authentications	Daniel Reev and Priya Shah
Confirmed data exfiltration	Not established
Confirmed customer-data compromise	Not established

2. Incident Overview

Field	Assessment
Incident	Executive account takeover
Affected user	Daniel Reev — CEO
Initial suspicious source	203.0.113.41 / Lagos, Nigeria
Successful authentication	10 Aug 2026 23:12 UTC
Post-authentication activity	Outlook Web and Azure Portal
Persistence / account manipulation	New authenticator registered on Pixel 6
Mailbox manipulation	Finance/invoice-focused inbox rule
Overall status	Confirmed compromise; response simulated

3. Investigation Objectives

• Confirm or refute compromise.

• Reconstruct the activity timeline.

• Identify persistence or security-control changes.

• Determine whether other accounts were targeted or accessed.

• Assess confirmed business impact.

• Develop repeatable Splunk detections.

4. Data Sources & Method

Source	Use	Volume
Sign-in telemetry	Authentication, IP, location, device, application and result	1,498

Source	Use	Volume
Audit telemetry	Account changes, security registration and mailbox actions	166
Splunk Enterprise	Search, correlation, timeline and detection testing	Primary

All telemetry is synthetic. Documentation-range IP addresses, fictional identities and lab data were used.

5. Initial Detection & Triage

Observation	Evidence	Assessment
Unusual geography	Lagos, Nigeria	Outside observed London baseline
Unusual device	Unknown Windows Device	Not in CEO baseline
Repeated failures	6 failures before success	Consistent with credential attack activity
Successful access	23:12 UTC from same source	Required audit correlation

6. Detailed Attack Timeline

UTC	Type	Source	Activity	Result	Detail
23:09	Sign-in	203.0.113.41	M365	Failure	50126
23:10	Sign-in	203.0.113.41	M365	Failure	50126
23:12	Sign-in	203.0.113.41	M365	Success	0
23:14	Sign-in	203.0.113.41	Outlook Web	Success	0
23:18	Audit	203.0.113.41	RegisterSecurityInfo	Success	Pixel 6
23:26	Sign-in	203.0.113.41	Azure Portal	Success	0
23:31	Audit	203.0.113.41	New-InboxRule	Success	RSS subscriptions

7. Attack Chain Assessment

Stage	Observed Evidence	Assessment
Credential attack	Repeated 50126 failures across users	Password-spray behavior observed
Initial access	Successful Daniel authentication from .41	Compromise supported by subsequent audit activity
Account manipulation	Pixel 6 authenticator registered	Potential continued access mechanism
Mailbox manipulation	Finance/invoice rule created	Potential concealment/monitoring
Scope	26 targeted; 2 successful	Environment-wide targeting

8. Persistence & Mailbox Manipulation

Time	Operation	Device	Action	Significance
23:18	RegisterSecurityInfo	Pixel 6	New authenticator	Potential continued access
23:31	New-InboxRule	Unknown Windows Device	Finance/invoice mail moved and marked read	Potential concealment

9. Scope Assessment

Metric	Result	Interpretation
Unique accounts targeted	26	Broad suspicious failed-authentication targeting
Successful suspicious authentications	2	Daniel Reev and Priya Shah
Other targeted accounts without observed success	24	No success observed in attacker-IP query
Suspicious source IPs	.41 / .42 / .43	Synthetic attack infrastructure

10. Business Impact Assessment

Confirmed / Observed	Not Established
CEO account compromise	Actual data exfiltration
Unauthorized security-information change	Financial transaction fraud
Mailbox rule affecting finance/invoice messages	Customer data theft
Potential exposure/disruption of finance communications	Customer account compromise

The report deliberately separates observed impact from outcomes that cannot be established from the available telemetry.

11. MITRE ATT&CK; Mapping

Technique	Name	Evidence
T1110.003	Password Spraying	Repeated failed authentication attempts against multiple users
T1098.005	Account Manipulation: Device Registration	New Pixel 6 authenticator registered after suspicious access
T1564.008	Email Hiding Rules	Inbox rule moved finance/invoice messages and marked them read

12. Containment & Response Plan

ID	Action	Status
C-01	Revoke active CEO sessions	Simulated
C-02	Reset CEO password / invalidate credential	Simulated
C-03	Remove unauthorized Pixel 6 authenticator	Simulated
C-04	Remove unauthorized inbox rule	Simulated
C-05	Block observed suspicious source IPs	Simulated

Because the dataset is static, historical creation of the inbox rule remains searchable. Splunk history alone cannot prove current mailbox state after simulated deletion.

13. Detection Engineering

Detection	Purpose	Logic
Password Spray	Distributed failed authentication	50126 + ≥3 failures/user + ≥3 users/source
Failure → Success	Possible credential compromise	≥3 failures + ≥1 success/source/user
New Security Registration	New authentication method	RegisterSecurityInfo event
New Inbox Rule	Mailbox manipulation	New-InboxRule + Create + Success

14. Key Splunk Queries

Password spraying

```
index=cloudora sourcetype=cloudora:signin | search result_code=50126 | stats count as failed_attempts by src_ip, user | where failed_attempts >= 3 | stats dc(user) as unique_users, sum(failed_attempts) as total_failures by src_ip | where unique_users >= 3
```

Failure → success

```
index=cloudora sourcetype=cloudora:signin | stats count(eval(result_code=50126)) as failed_attempts,
count(eval(result="Success")) as successful_attempts by src_ip, user | where failed_attempts >= 3 AND successful_attempts >= 1
```

New registration

```
index=cloudora sourcetype=cloudora:audit | search operation="RegisterSecurityInfo" | stats count by user, src_ip, device, target, action, result
```

Inbox rule

```
index=cloudora sourcetype=cloudora:audit | search operation="New-InboxRule" action="Create" result="Success" | table _time, user, src_ip, operation, target, device, action, result, details
```

15. Limitations

• Synthetic telemetry only; no production environment was involved.

• No full endpoint, packet, email-content or DLP telemetry was available.

• Successful authentication alone does not prove malicious intent; correlation supports the assessment.

• Historical Splunk events cannot prove current mailbox state after response.

• No claim of exfiltration, fraud or customer compromise is made.

16. Final Incident Assessment

FINAL STATUS: CONFIRMED ACCOUNT COMPROMISE

The evidence supports a confirmed compromise of the Cloudora CEO account. The observed chain is password-spray activity → successful suspicious authentication → Outlook Web and Azure Portal access → new authenticator registration → finance/invoice-focused inbox rule.

Twenty-six unique accounts were targeted. Two accounts showed successful authentication from suspicious infrastructure. Account and mailbox manipulation are confirmed by available telemetry; broader outcomes such as data exfiltration, financial fraud and customer-data compromise remain unestablished.

Appendix — Evidence Package

Artifact	Purpose
SOC Investigation Evidence Log	Preserves investigation questions, SPL, evidence and analyst interpretation
Splunk Dashboard	Visualizes authentication, suspicious sources, scope and timeline
Synthetic Sign-in Dataset	Authentication telemetry used for investigation
Synthetic Audit Dataset	Account and mailbox activity used for correlation
Incident Report	Final analyst-facing assessment